

JEE MAIN REVISION MASTER: STRUCTURE OF ATOM SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 219 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Bohr's Orbitals & Electronic Transitions	124 PYQs	Bohr Model
Very High	Quantum Numbers & Electronic Configuration	69 PYQs	Quantum Numbers
Very High	Heisenberg Uncertainty & De Broglie Wavelength	34 PYQs	Wave Mechanics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Bohr's Orbitals & Electronic Transitions	$r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$, $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$, $\frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$
Quantum Numbers & Electronic Configuration	$\text{Radial nodes} = n - l - 1$, $\text{Angular nodes} = l$, $\text{Total nodes} = n - 1$, $\mu_B = \frac{e h}{4 \pi m}$
Heisenberg Uncertainty & De Broglie Wavelength	$\Delta x \Delta p \geq \frac{h}{4\pi}$, $\Delta x \Delta v \geq \frac{h}{4\pi m}$, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Bohr's Orbitals & Electronic Transitions (Very High Yield — 124 PYQs)

Core Principle: Electronic energy levels in hydrogenic species, wave number $\bar{\nu}$, and emission spectral series.

■ **Caution / JEE Pitfall:** Rydberg formula requires multiplication by Z^2 for single-electron ions like He^+ ($Z=2$) and Li^{2+} ($Z=3$).

■ **Speed Trick:** Limiting line (series limit) of any spectral series occurs when $n_2 = \infty \implies \bar{\nu}_{(\text{limit})} = (R Z^2 / n_1^2)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The Bohr radius of a hydrogen like species is 70.53 pm. The species and the stationary state (n) are respectively (Given : Hydrogen atom Bo..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Consider the following spectral lines for atomic hydrogen : A. First line of Paschen series B. Second line of Balmer series C. Third line of..." [Ans: C]
- [2026 Evening Shift] Q#1 (Numerical): "The surface of sodium metal is irradiated with radiation of wavelength λ nm. The kinetic energy of ejected electrons is 2.8×10^{-2} ..." [Ans: 5]

3.2 Quantum Numbers & Electronic Configuration (Very High Yield — 69 PYQs)

Core Principle: Orbital shapes, quantum numbers (n, l, m_l, m_s), nodal surfaces, and anomalous configurations (Cr: $3d^5 4s^1$, Cu: $3d^{10} 4s^1$).

■ **Caution / JEE Pitfall:** Spin-only magnetic moment $\mu = \sqrt{n(n+2)}$ BM depends on n , which is the number of UNPAIRED electrons, not total electrons!

■ **Speed Trick:** Degeneracy of hydrogen atom at energy level n is n^2 (without spin) or $2n^2$ (including spin).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Arrange the following atomic orbitals of multi electron atoms in order of increasing energy. A. $n=3, l=2, m=+1$ B. $n=4, l=0, m=0$ C. $n=6, l=1, m=0$..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Which of the following is correct set of 4 quantum numbers of 19th electron in Chromium (Atomic number =24) in accordance with..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Figure 1. electron probability density for 2s orbital Figure 2. wave function for 2s orbital Which of the following point in Figure 2 most ..." [Ans: N/A]

3.3 Heisenberg Uncertainty & De Broglie Wavelength (Very High Yield — 34 PYQs)

Core Principle: Limits of simultaneous precision in position and momentum, and dual wave nature of micro-particles.

■ **Caution / JEE Pitfall:** If uncertainty in position and momentum are equal ($\Delta x = \Delta p$), then $\Delta v = \frac{1}{2m} \sqrt{\frac{h}{\pi}}$.

■ **Speed Trick:** De Broglie wavelength of an electron accelerated through potential V volts: $\lambda \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "Identify the correct statements from the following : A. Heisenberg uncertainty principle is applicable to electrons. B. The size of $2p_x$..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (Numerical): "Two positively charged particles m_1 and m_2 have been accelerated across the same potential difference of 200 keV as shown below. [...]" [Ans: 2]
- [2025 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement (I) : It is impossible to specify simultaneously with arbitrary precision, both the linear moment..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

Consider the following spectral lines for atomic hydrogen : A. First line of Paschen series B. Second line of Balmer series C. Third line of Paschen series D. Fourth line of Brackett series The correct arrangement of the above lines in ascending order of energy is :

- (A) < C < A < B
- (B) < B < C < D
- (C) < A < C < B [Correct]
- (D) < D < B < A

Correct Answer: C

Solution Analysis: For hydrogen spectral lines (emission), $E = \frac{hc}{\lambda} \Rightarrow E \propto \frac{1}{\lambda}$ Using Rydberg formula (NCERT): $\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ ($n_2 > n_1$) So photon energy is proportional to: $\Delta = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ Now identify each line: Paschen series : $n_1=3$ A (first line) : $n_2=4 \rightarrow 3$ C (third line) : $n_2=6 \rightarrow 3$ Balmer series : $n_1=2$ B (second line) : $n_2=4 \rightarrow 2$ Brackett series : $n_1=4$ D (fourth line) : $n_2=8 \rightarrow 4$ Compute Δ for each: D: $\frac{1}{16} - \frac{1}{64} = \frac{3}{64}$

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Evening Shift (Q#2)

Two positively charged particles m_1 and m_2 have been accelerated across the same potential difference of 200 keV as shown below. [Given mass of $m_1=1$ amu and $m_2=4$ amu] The deBroglie wavelength of m_1 will be x times of m_2 . The value of x is _____ (nearest integer)

Correct Answer: 2

Solution Analysis: $\lambda = \frac{h}{\sqrt{2mK.E.}}$ Here, both particles are accelerated through the same potential difference of 200 keV, so they gain the same kinetic energy (K.E.). So, in the formula, h and K.E. are the same for both particles. Only the mass m is different. Therefore, $\lambda \propto \frac{1}{\sqrt{m}}$ Now, $\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{4}{1}} = 2$ So $x=2$.

Benchmark Problem 3: JEE Main 2026 (Online) 6th April Morning Shift (Q#3)

Match the LIST-I with LIST-II List-I Orbital List-II Radial nodes and nodal plane A. 2s I. 1 Radial node + two nodal planes B. 3s II. 1 Radial node + one nodal plane C. 3p III. 2 Radial nodes + No nodal plane D. 4d IV. 1 Radial node + No nodal plane Choose the correct answer from the options given below:

- (A) -IV, B-I, C-III, D-II
- (B) -IV, B-II, C-III, D-I
- (C) -III, B-I, C-IV, D-II
- (D) -IV, B-III, C-II, D-I **[Correct]**

Correct Answer: D

Solution Analysis: A node is a region in space where the probability of finding an electron is zero. There are two types of nodes: Radial Nodes (or Spherical Nodes): These are spherical surfaces where the probability of finding an electron is zero. Angular Nodes (or Nodal Planes): These are planes (or cones for d-orbitals) passing through the nucleus where the probability of finding an electron is zero. To find the number of these nodes for any given orbital, we use two simple formulas based on the principal quantum number (n) and the azimuthal quantum number (l). Number of radial nodes = $n-l-1$ Number